

LUIS HENRICH-BANDIS

Solo Founder & Applied AI Engineer

Cologne, Germany · +49 174 3856722 · hi@luishenrich.com

studypdf.net · luishenrich.com · github.com/luishenrich · linkedin.com/in/luishenrich

SUMMARY

Native-German technical founder. Built StudyPDF to 85,000 users and \$1.5k MRR using multi-agent orchestration over Anthropic and OpenAI APIs. Profitable from day one with 99% organic growth. Comfortable moving between prototyping, debugging, evaluation design, and shipping. Currently deepening ML foundations through Deep Learning coursework at RWTH Aachen.

EXPERIENCE

Founder & Sole Engineer · *StudyPDF, studypdf.net*

10/2023 – Present

- **Built and scaled to 85,000 users** organically. DAU grew from ~100 to 600+ over the last 5 months, generating \$1.5k MRR with 99% organic growth.
- **Designed a two-stage extraction pipeline:** one expensive reasoning-model pass extracts topic/concept structure per document; every downstream feature (flashcards, exams, mind maps, chat) reuses the same structured output. Significantly reduces per-feature inference cost while improving consistency.
- **Debug model behaviour in production daily:** diagnose failure modes across 150k+ processed documents, distinguish model-level from workflow-level issues, iterate on prompts and agent designs.
- **Run continuous evaluation loops** via A/B tests against real user behaviour; currently building a reproducible eval system to score concept extraction against ground truth.
- **Stack:** TypeScript, Next.js, PostgreSQL, Pinecone, Anthropic/OpenAI APIs, Cloudflare R2, Claude Code + PostHog MCP for daily ops.
- **Recent writing:** "How I run an 85,000-user AI study platform solo", at luishenrich.com/blog/85k-users-solo

AI Engineer Intern · *BMW Group, Munich*

10/2024 – 03/2025

- **Shipped an internal RAG chat app** on a self-hosted local LLM enabling Virtual Commissioning engineers to query production planning documentation conversationally.
- **Built an LLM tool generating state diagrams** and technical docs from product specifications, accelerating engineering documentation workflows.
- Rated 'exceeds requirements significantly' (top tier) by supervisor Alexander Kuhn. Quote: "work pace consistently well above our expectations." Recommendation letter available on request.

EDUCATION

M.Sc. Business Administration & Mechanical Engineering · *RWTH Aachen*

10/2025 – Present

GPA 1.7 (German scale, "good"). Specialization: Production Technology & Operations Research. ML coursework: Fundamentals of ML, Deep Learning in Economics, ML for Business Analytics. Production Management I: 1.0; Industrial Statistics: 1.3.

B.Sc. Business Administration & Mechanical Engineering · *RWTH Aachen*

10/2021 – 09/2025

Final GPA 1.8 ("good"), completed in standard period. Bachelor's thesis 1.0 ("very good") on business potential of a European energy data platform.

CS50x: Introduction to Computer Science · *Harvard University (online), 2023*. Completed 10 problem sets, 9 labs, and a final project covering C, Python, SQL, and web development.

SKILLS

Languages: German (native), English (professional working proficiency).

Applied AI: Multi-agent orchestration, evaluation design, production debugging, failure-mode analysis, cost & latency optimization, RAG, prompt engineering, Anthropic & OpenAI APIs, Claude Code, PostHog MCP, Pinecone.

Engineering: TypeScript, Node.js, Next.js, React, Python, PostgreSQL, Cloudflare R2, Vercel, Git.